What might the midterm ask?

Evidence from Quiz 1, checked against all seven supplied lecture decks

The safest prediction is the style: write a formula, use it on small numbers, and explain what the result means. Quiz 1 used one four-mark explanation of GAN losses and one four-mark Inception Score calculation.

These are tentative question archetypes, not advance knowledge of the paper. The mock marks and timing below are our practice design.

Priority / evidence	Plausible question wording
First: directly seen in Quiz 1	Write saturating/non-saturating GAN losses. Explain or derive what happens when D(G(z)) is near zero.
First: directly seen in Quiz 1	Given classifier probabilities, compute the marginal, each KL divergence, the mean and the Inception Score.
First: worked in the lectures	Compute a diagonal-Gaussian VAE KL, sample by reparameterization, and combine reconstruction and KL losses.
First: worked in the lectures	Compute self-attention or cross-attention; apply a causal mask; calculate token cross-entropy.
First: worked in the lectures	Compute forward diffusion from the cumulative schedule, or a reverse mean and noisy sample.
Next: worked in the lectures	Calculate diagonal FID and explain its differences from IS and generative precision/recall.
Next: lecture concepts	Explain mode collapse, AE versus VAE, cGAN conditioning, StackGAN stages, or positional encoding.
Backup: more demanding variants	Derive VAE KL, the ELBO, aggregate diffusion noise, the reverse mean, or the optimal discriminator.

Priority measures usefulness for preparation. It does not estimate the probability of appearing. A single quiz cannot establish the actual exam length, topic weights, or question count.

How to use this practice pack

Questions first, solutions later; focus on the reasoning you can reproduce

A practical route for tonight

Step	Time	Action
1	90 min	Attempt Q1-Q8 on paper with the solutions covered.
2	45-60 min	Review worked solutions. Redo every part you missed from a blank page.
3	30-45 min	Use the alternate questions to repair weak topics; do not attempt all by default.
4	10 min	Explain each model and the common traps aloud.

If time is tight, start with Q3 (GAN), Q4 (metrics), Q2 (VAE), Q5-Q6 (diffusion), then Q7-Q8 (transformers). Do not omit a topic solely because it was absent from the quiz.

Assumptions for this original mock

60 marks, suggested 90 minutes; all eight questions are compulsory for practice. This is not a claim about your instructor's format. A calculator is useful. Use natural logs, radians for trigonometry, and retain precision until the final answer. Show formulas, substitution, intermediate results, and interpretation.

Source map: PDF page positions, including cover slides

Source	Used to design
Quiz 1 MSDS solution.pdf, pp. 1-3	GAN explanation + three-image IS arithmetic.
Lecture1 Introduction.pdf, pp. 15-29, 37-41	Joint/conditional modeling, Bayes, quality and coverage.
Lecture 2 AE and VAE v2.pdf, pp. 5-19, 36-50	Reconstruction, latent codes, KL and reparameterization.
Lecture GANs Part 1.pdf, pp. 28-47; Part 2, pp. 1-18, 20-37, 53-95	Losses, failure modes, conditional generation, IS/FID and precision/recall.
Lecture Diffusion Models Part 1.pdf, pp. 14-31; Part 2, pp. 19-29, 35-46	Variance, aggregate noising, reverse mean, objectives and sampling.
Transformer.pdf, pp. 20-22, 30-43, 55-58, 62-67	Attention, masks, position encoding, cross-entropy.

Scope follows the seven supplied lecture decks, as requested. Topics named only in the semester outline are not treated as fully taught units. Question wording and numerical values in this pack are original.

Mock paper: Q1 and Q2

14 marks / suggested 21 minutes

Q1. Generative modeling and Bayes [4 marks]

An email has a spam prior of 0.30. The word "win" appears in 80% of spam emails and 10% of non-spam emails.

(a) Calculate P(spam | win). Show the normalizing denominator. [2]
(b) Distinguish modeling p(x,y) from modeling p(y|x). Is a text-conditioned image generator a discriminative classifier simply because it has a condition? Explain. [2]

Q2. AE reconstruction and VAE sampling [10 marks]

A two-pixel input and its autoencoder reconstruction are given below. The VAE encoder separately returns a two-dimensional mean and log-variance:

$$x = (1, 0), \quad \hat{x} = (0.8, 0.1), \quad \mu = (1, -1)$$

$$\ell = \ln \sigma^2 = (\ln 4, 0), \quad \varepsilon = (0.5, -1)$$

(a) Calculate the reconstruction MSE, averaged over the two pixels. [2]
(b) Convert log-variance into variance and standard deviation. Calculate a latent sample z using reparameterization. [2]
(c) Calculate KL(q(z|x) || N(0,I)), summed over latent coordinates. [3]
(d) Let the toy beta-VAE objective be L = MSE + 0.2 KL. Compute L. [1]
(e) Explain why reparameterization permits gradient training, and why a plain AE does not automatically give a reliable standard-normal sampling space. [2]

Use the stated MSE and summed KL convention. No batch averaging or Gaussian decoder constant is required in this question.

Mock paper: Q3 and Q4

18 marks / suggested 27 minutes

Q3. GAN losses and learning signals [10 marks]

Let d = D(G(z)) = sigmoid(a), where a is the discriminator logit for one generated sample.

(a) Write the GAN minimax value function and the discriminator loss when implemented by minimization. Write the saturating and non-saturating generator losses. [3]
(b) Derive the two generator derivatives with respect to a. At d = 0.02, calculate both losses and both logit derivatives. Explain why the non-saturating version is useful early in training. [4]
(c) At one location, p_data(x) = 0.30 and p_g(x) = 0.10. Calculate the optimal D*(x). [1]
(d) The generator produces convincing faces, but nearly all show the same identity. Name and explain the failure, and propose one response without claiming it guarantees a cure. [2]

Q4. Inception Score and FID [8 marks]

An Inception classifier gives the following class probabilities for three generated images:

Image	Class 1	Class 2
1	0.90	0.10
2	0.20	0.80
3	0.40	0.60

(a) Calculate the Inception Score. Show the marginal, each KL, their mean, and the exponentiation. [4]
(b) Real and generated features have means (0,1) and (1,3), and diagonal covariances diag(1,4) and diag(4,9). Calculate FID. [3]
(c) Every generated image receives the same confident class prediction (1,0). What IS results, and why? [1]

Mock paper: Q5 and Q6

14 marks / suggested 21 minutes

Q5. Forward diffusion and aggregate noise [8 marks]

Use a two-step DDPM schedule with beta_1 = 0.10 and beta_2 = 0.20. The clean vector is x_0 = (1,-1).

(a) Calculate alpha_1, alpha_2 and cumulative alpha_bar_2. [1]
(b) Use the aggregate standard-normal noise realization epsilon = (0,1) to calculate x_2 directly from x_0. [2]
(c) Write the conditional mean and covariance of q(x_2 | x_0). Treat x_0 as fixed. [2]
(d) Expand two sequential forward steps with independent epsilon_1 and epsilon_2, each N(0,I). Show that the combined noise has covariance (1 - alpha_bar_2)I and can be represented with one standard-normal vector. [3]

The aggregate epsilon in (b) is not automatically equal to the noise added at step 2. Part (d) explains the distinction.

Q6. A reverse DDPM step [6 marks]

At t = 2 use the same schedule as Q5, but a different current sample x_t = (0.5,-0.5). The network predicts epsilon_theta = (0.2,-0.1). Use the following mean formula:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta(x_t, t)\right)$$

(a) Calculate the reverse mean. [2]
(b) For this question, choose sigma_t squared = beta_tilde_t = beta_t (1-alpha_bar_(t-1))/(1-alpha_bar_t). Calculate the variance and standard deviation, then sample x_(t-1) using z = (1,-1). [2]
(c) Why can the exact posterior conditioned on x_0 not simply be used during generation? What is normally done with the added noise at the final t = 1 step? [2]

Mock paper: Q7 and Q8

14 marks / suggested 21 minutes

Q7. Self-attention, masking and shapes [10 marks]

Use row vectors. The three token embeddings are x_1=(1,0), x_2=(0,1), x_3=(1,1). The query, key and value projection matrices are all I_2. There are no position vectors; d_k=d_v=2. Find the single-head weighted-value output before any output projection, residual addition or normalization.

(a) For token 2, calculate q_2, the three scaled dot-product scores, row-softmax weights and attention output. [5]
(b) Treat the same tokens as shifted decoder inputs. Apply causal masking for position 2 and recompute its weights and output. Which positions are allowed? [3]
(c) Cross-attention has 2 target positions and 3 source positions with d_k=d_v=2. State the shapes of Q, K, V, QK^T, and the attention output; identify where Q, K and V come from. [2]

Apply softmax across keys in each query row. A masked score becomes negative infinity before softmax.

Q8. Position encoding and token loss [4 marks]

(a) Compute all four sinusoidal positional-encoding components for pos = 3 and d_model = 4. Use zero-based indices and radians. [2]

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

$$PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

(b) At three non-padding target positions, the predicted probabilities of the correct tokens are 0.80, 0.50 and 0.25. Calculate mean token cross-entropy in nats. [2]

End of mock questions. Stop here until you have attempted the paper.

Solution Q1: classify from a joint model

4 marks / suggested marking points

1. Normalize the two joint scores [2]

Let S mean spam, H mean non-spam, and W mean that "win" is present. The numerator is the probability of both spam and the word. The denominator includes every way the word could appear.

$$P(S, W) = 0.30(0.80) = 0.24, \quad P(H, W) = 0.70(0.10) = 0.07$$

$$P(W) = 0.24 + 0.07 = 0.31$$

$$P(S \mid W) = \frac{0.24}{0.31} = 0.774194 \approx 77.42\%$$

Check: P(H|W)=0.07/0.31=0.225806, so the two posteriors sum to 1. With equal error costs and a 0.5 threshold, this email would be classified as spam.

Marking: one point for the joint terms and denominator; one for the normalized posterior.

2. State what is modeled [2]

A model of p(x,y) describes the joint distribution of inputs and labels. It can obtain p(y|x) by normalizing the joint over labels. A model of p(y|x) alone describes labels given inputs; it does not supply a distribution for generating new inputs x.

A text-to-image model is conditional generation: it samples an image x given text c, representing p(x|c) or an implicit conditional sampler. The existence of a condition does not turn generation into classification.

Common error: 0.80 is P(win|spam), not P(spam|win). Reversing the conditioning skips both the prior and the normalization.

Solution Q2: reconstruction plus regularity

10 marks / MSE over pixels, KL summed over latent dimensions

1. Reconstruction [2]

$$\text{MSE} = \frac{(0.8-1)^2 + (0.1-0)^2}{2} = \frac{0.05}{2} = 0.025$$

2. Turn log-variance into a sample [2]

$$\sigma^2 = e^{\ell} = (4, 1), \quad \sigma = e^{\ell/2} = (2, 1)$$

$$z = \mu + \sigma \odot \varepsilon = (1, -1) + (2, 1) \odot (0.5, -1) = (2, -2)$$

The multiplication is coordinate by coordinate. A variance of 4 means a standard deviation of 2, so 2 is the sampling multiplier.

3. KL to a standard-normal prior [3]

$$\text{KL} = \tfrac{1}{2}\sum_j \left(\mu_j^2 + \sigma_j^2 - 1 - \ln \sigma_j^2\right)$$

$$\text{KL} = \tfrac{1}{2}[(1 + 4 - 1 - \ln 4) + (1 + 1 - 1 - 0)]$$

$$\text{KL} = \tfrac{1}{2}(5 - \ln 4) = 1.806853$$

The two coordinate contributions are 1.306853 and 0.500000. Marking: correct formula, substitutions and total.

4. Combine the stated terms [1]

$$L = 0.025 + 0.2(1.806852819) = 0.386371$$

5. Explain the mechanism [2]

Reparameterization places randomness in an auxiliary epsilon independent of the encoder parameters. For a fixed sampled epsilon, z=mu+sigma*epsilon is differentiable in mu and sigma, so the reconstruction gradient reaches the encoder. A plain AE learns codes for reconstruction but does not enforce a standard-normal prior; random normal codes can land in poorly trained latent regions.

Do not multiply by log-variance when sampling, average this KL over two coordinates, or assume the toy MSE objective is the complete likelihood ELBO without decoder assumptions.

Solution Q3: which derivative vanishes?

10 marks / distinguish discriminator output d from logit a

1. Objectives and optimization directions [3]

$$\min_G \max_D V = \mathbb{E}_{x \sim p_{data}}[\ln D(x)] + \mathbb{E}_{z \sim p_z}[\ln(1 - D(G(z)))]$$

$$L_D = -V, \quad L_G^{sat} = \mathbb{E}_z[\ln(1-d)], \quad L_G^{NS} = -\mathbb{E}_z[\ln d]$$

The discriminator minimizes the negative of its maximization objective. The generator minimizes either stated generator loss; the real-data term does not depend on G.

2. Apply the sigmoid chain rule [4]

$$\frac{\partial d}{\partial a} = d(1-d)$$

$$\frac{\partial L^{sat}}{\partial a} = \frac{-1}{1-d} d(1-d) = -d$$

$$\frac{\partial L^{NS}}{\partial a} = \frac{-1}{d} d(1-d) = d - 1$$

At d = 0.02	Saturating	Non-saturating
Loss	ln(0.98) = -0.020203	-ln(0.02) = 3.912023
Derivative w.r.t. a	-0.020000	-0.980000

As d approaches 0, the logit gradient of the saturating loss approaches 0; the non-saturating gradient approaches -1. Gradient descent raises the logit and hence the fake sample's realness score. This preserves a stronger local training signal when G is initially poor. The complete parameter gradient also contains the remaining network derivatives.

3. Optimal discriminator and collapse [1 + 2]

$$D^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_g(x)} = \frac{0.30}{0.40} = 0.75$$

The face example is mode collapse: convincing outputs with poor coverage or diversity. Minibatch-aware discrimination is one possible response because it can expose repeated samples. Stabilizing training can help, but no single intervention guarantees recovery of every mode.

Trap: derivatives w.r.t. d are -1/(1-d) and -1/d. They are not -d and d-1. Also, a loss value close to zero need not mean this generator is doing well.

Solution Q4: evaluate confidence and coverage

8 marks / IS higher is better; FID lower is better, within matched protocols

1. Marginal label distribution [part of 4]

$$p(y) = \frac{(0.9, 0.1) + (0.2, 0.8) + (0.4, 0.6)}{3} = (0.5, 0.5)$$

2. Conditional-to-marginal KL for each image

$$\mathrm{KL}_i = \sum_k p(k \mid x_i) \ln \frac{p(k \mid x_i)}{p(k)}$$

Image	Substitution with natural logs	KL (nats)
1	0.9 ln(1.8) + 0.1 ln(0.2)	0.368064
2	0.2 ln(0.4) + 0.8 ln(1.6)	0.192745
3	0.4 ln(0.8) + 0.6 ln(1.2)	0.020136

$$\overline{\mathrm{KL}} = 0.193648159, \quad IS = \exp(0.193648159) = 1.213669$$

Show all four stages: marginal, per-image KL, average and exponentiation. Here IS lies between 1 and the two-class ceiling 2. Confident predictions and a varied marginal label distribution both contribute.

3. Diagonal FID [3]

$$FID = \|\mu_r - \mu_g\|^2 + \sum_j (v_{r,j} + v_{g,j} - 2\sqrt{v_{r,j} v_{g,j}})$$

$$\|\mu_r - \mu_g\|^2 = (0 - 1)^2 + (1 - 3)^2 = 5$$

$$\text{spread} = (1 + 4 - 2\sqrt{4}) + (4 + 9 - 2\sqrt{36}) = 1 + 1 = 2$$

$$FID = 5 + 2 = 7$$

4. All predictions in one class [1]

The marginal is also (1,0), so every conditional equals the marginal. Every KL is 0 and IS=exp(0)=1. The 0 ln(0/q) contribution is interpreted by its limiting value 0; absent classes contribute no mass. Confident but single-class generation does not earn a high IS.

IS uses no real reference distribution and can miss within-class collapse. FID uses feature means and covariances; even FID=0 does not prove identical full image distributions or rule out memorization.

Solution Q5: one jump to any noise level

8 marks / the clean input is fixed in this conditional distribution

1. Schedule and direct sample [1 + 2]

$$\alpha_1 = 0.9, \quad \alpha_2 = 0.8, \quad \bar{\alpha}_2 = 0.9(0.8) = 0.72$$

$$x_2 = \sqrt{0.72}\, x_0 + \sqrt{0.28}\, \varepsilon$$

$$x_2 = (0.848528,\ -0.848528) + (0, 0.529150) = (0.848528,\ -0.319378)$$

2. Conditional distribution [2]

$$q(x_2 \mid x_0) = \mathcal{N}(\sqrt{0.72}\, x_0, 0.28I)$$

$$\mathbb{E}[x_2 \mid x_0] = (0.848528,\ -0.848528), \quad \mathrm{Cov}(x_2 \mid x_0) = 0.28I$$

The sample in part (b) differs from the conditional mean because it contains a particular draw of noise. The conditional covariance describes variability across repeated noise draws, not the squared entries of that one sample.

3. Expand the two steps [3]

$$x_1 = \sqrt{\alpha_1}\, x_0 + \sqrt{\beta_1}\, \varepsilon_1$$

$$x_2 = \sqrt{\alpha_2\alpha_1}\, x_0 + \sqrt{\alpha_2\beta_1}\, \varepsilon_1 + \sqrt{\beta_2}\, \varepsilon_2$$

The independent Gaussian noise terms add to another zero-mean Gaussian. Covariance coefficients are squared:

$$\mathrm{Cov}(\text{noise}) = (\alpha_2\beta_1 + \beta_2)I = (0.08 + 0.20)I = 0.28I$$

$$\alpha_2\beta_1 + \beta_2 = \alpha_2(1 - \alpha_1) + (1 - \alpha_2) = 1 - \alpha_1\alpha_2$$

$$\varepsilon = \frac{\sqrt{0.08}\,\varepsilon_1 + \sqrt{0.20}\,\varepsilon_2}{\sqrt{0.28}} \sim \mathcal{N}(0, I)$$

The aggregate epsilon is a normalized combination of earlier noises. Do not add standard deviations, or substitute alpha_2=0.8 where cumulative alpha_bar_2=0.72 is needed.

Solution Q6: predict noise, then denoise

6 marks / posterior-variance choice specified by the question

1. Compute the reverse mean [2]

$$\frac{\beta_2}{\sqrt{1-\bar{\alpha}_2}} = \frac{0.2}{\sqrt{0.28}} = 0.377964473$$

$$\mu_\theta = \frac{(0.5,\ -0.5) - 0.377964473(0.2,\ -0.1)}{\sqrt{0.8}}$$

$$\mu_\theta = (0.474501569,\ -0.516759282)$$

The model predicts the noise direction. The mean subtracts a schedule-scaled amount of this predicted noise, then rescales the result. It is not simply x_t minus epsilon_theta.

2. Variance, standard deviation and sample [2]

$$\tilde{\beta}_2 = \frac{0.2(1-0.9)}{1-0.72} = \frac{0.02}{0.28} = \frac{1}{14} = 0.071428571$$

$$\sigma_2 = \sqrt{\tilde{\beta}_2} = 0.267261242$$

$$x_1 = \mu_\theta + \sigma_2 z$$

$$x_1 = (0.474501569,\ -0.516759282) + 0.267261242(1,\ -1)$$

$$x_1 = (0.741762811,\ -0.784020524)$$

The Gaussian draw is scaled by the standard deviation, not the variance. A sampled reverse step need not move each coordinate closer to the clean image.

3. What is known at generation time? [2]

The exact Gaussian q(x_(t-1)|x_t,x_0) requires the clean x_0, which is unavailable when generating a new image. Instead, the learned network estimates noise (or a related parameterization) from x_t and t, allowing a learned reverse transition.

At the final t=1 step, use the reverse mean without adding fresh Gaussian noise. In this posterior-variance convention, the formula also gives beta_tilde_1=0 because alpha_bar_0=1.

This Gaussian claim is conditioned on both x_t and x_0. The marginal reverse conditional q(x_(t-1)|x_t) is not generally exactly Gaussian.

Solution Q7: attention is a weighted sum

10 marks / row-vector convention used throughout

1. Project and score token 2 [5]

$$q_2 = x_2 I = (0, 1), \quad K = V = [\,(1, 0);\ (0, 1);\ (1, 1)\,]$$

$$\frac{q_2 K^T}{\sqrt{d_k}} = \frac{(0, 1, 1)}{\sqrt{2}} = (0, 0.707107, 0.707107)$$

$$a_j = \frac{e^{s_j}}{\sum_k e^{s_k}}, \quad a = (0.197776, 0.401112, 0.401112)$$

$$o_2 = a_1(1, 0) + a_2(0, 1) + a_3(1, 1) = (0.598888, 0.802224)$$

The weights sum to 1. Their job is to mix value vectors; they are not the final output features. Marks: query/keys and raw products, scaling, softmax, weighted sum, final vector.

2. Causal decoder position 2 [3]

Position 2 can attend to positions 1 and 2, including its own shifted input, but not future position 3. Set the third score to negative infinity before softmax:

$$s_{masked} = (0, 0.707107, -\infty)$$

$$a_{masked} = (0.330238, 0.669762, 0)$$

$$o_{2,\,masked} = 0.330238(1, 0) + 0.669762(0, 1) = (0.330238, 0.669762)$$

Recompute the denominator over the permitted keys. Setting the old third weight to zero without renormalizing leaves the remaining weights summing to less than 1.

3. Cross-attention dimensions and sources [2]

Quantity	Shape	Comes from
Q	2 x 2	Decoder representations at target positions.
K and V	3 x 2 each	Encoder outputs at source positions.
QK^T / attention weights	2 x 3	Every target query scores all source keys.
Output = weights times V	2 x 2	One value-weighted vector per target query.

A source padding mask can still apply to cross-attention. The decoder causal mask prevents looking ahead in the target sequence; it is not normally a triangle over source positions.

Solution Q8: order and token probabilities

4 marks / zero-based components; trig arguments in radians

1. Compute the two frequency pairs [2]

There are two pairs for a four-dimensional encoding: i=0 supplies components 0 and 1, and i=1 supplies components 2 and 3.

Pair	Denominator	Angle	Components
i = 0	10000^(0/4) = 1	3/1 = 3	sin(3), cos(3)
i = 1	10000^(2/4) = 100	3/100 = 0.03	sin(0.03), cos(0.03)

$$PE(3) = (0.141120,\ -0.989992, 0.029996, 0.999550)$$

Position vectors are added to token embeddings so the network can use sequence order. Different frequencies encode changes over different position scales. These formulas define encodings at unseen positions, but do not guarantee that a trained model will generalize accurately to arbitrary lengths.

2. Average the negative log of the correct tokens [2]

$$L_{CE} = -\tfrac{1}{3}[\ln(0.8) + \ln(0.5) + \ln(0.25)]$$

$$L_{CE} = -\tfrac{1}{3}\ln(0.1) = 0.767528 \text{ nats/token}$$

Correct-token probability	Negative natural log
0.80	0.223144
0.50	0.693147
0.25	1.386294
Sum / mean	2.302585 / 0.767528

For a one-hot target, only the probability assigned to the correct token contributes at that position. Do not choose the largest probability unless it happens to belong to the correct token. Padding positions are excluded here by the question.

Report whether the sentence loss is summed or averaged. Both conventions exist; this question explicitly asks for the mean.

Alternate A-B: VAE derivations

Optional: use this page if the numerical VAE question feels easy

A. Derive the scalar Gaussian KL [suggested 6 marks]

Question: derive KL(N(mu,v) || N(0,1)) from E_q[ln q(z) - ln p(z)], where v>0 is a variance. Extend the result to a diagonal multivariate Gaussian.

$$\ln q(z) - \ln p(z) = -\tfrac{1}{2}\ln v - \frac{(z-\mu)^2}{2v} + \frac{z^2}{2}$$

Under q, E[(z-mu)^2]=v and E[z^2]=v+mu^2. Substitute these two expectations:

$$\mathrm{KL} = -\tfrac{1}{2}\ln v - \tfrac{1}{2} + \tfrac{1}{2}(v+\mu^2) = \tfrac{1}{2}(\mu^2 + v - 1 - \ln v)$$

Independent diagonal coordinates factorize the densities, so the log ratio and its expectation sum over coordinates. The result is one half the sum of mu_j squared + v_j - 1 - ln(v_j). It is zero at mu=0,v=1 and is never negative.

Suggested marks: log ratio 2; correct expectations 2; closed form and diagonal extension 2.

B. Why is the ELBO a lower bound? [suggested 6 marks]

Question: express ln p_theta(x) as an ELBO plus a nonnegative KL. Identify reconstruction and prior regularization, and explain the minimization sign.

$$\mathrm{ELBO} = \mathbb{E}_{q_\phi(z\mid x)}[\ln p_\theta(x\mid z)] - \mathrm{KL}(q_\phi(z\mid x)\|p(z))$$

$$\ln p_\theta(x) = \mathrm{ELBO} + \mathrm{KL}(q_\phi(z\mid x)\|p_\theta(z\mid x))$$

The second KL is nonnegative, so ELBO cannot exceed the log evidence. Maximizing ELBO balances a good expected log-likelihood reconstruction with keeping the encoder distribution near the prior. Equivalently minimize negative expected log-likelihood plus prior KL.

The identity follows by substituting p_theta(z|x)=p_theta(x|z)p(z)/p_theta(x) into the posterior KL and rearranging. The posterior KL is the gap to the bound; the prior KL is the regularizer inside it. They are different terms.

MSE is proportional to Gaussian negative log-likelihood only after specifying a decoder variance and reduction. A beta-weighted loss is generally a modified objective, not automatically the standard ELBO.

Alternate C-D: gradients and dimensions

Optional: short arithmetic extensions to reconstruction and DCGAN

C. One tiny AE gradient update [suggested 6 marks]

Question: a linear tied-weight AE has input x=1, encoder z=wx and decoder x_hat=wz. At w=0.5, use L=(x_hat-x)^2 and learning rate 0.1. Find the forward values, dL/dw and one gradient-descent update. Recompute the loss.

$$z = 0.5, \quad \hat{x} = w^2 = 0.25, \quad L = (0.25 - 1)^2 = 0.5625$$

$$\frac{dL}{dw} = 2(w^2 - 1)(2w) = 4w(w^2 - 1) = -1.5$$

$$w_{new} = 0.5 - 0.1(-1.5) = 0.65$$

$$\hat{x}_{new} = 0.65^2 = 0.4225, \quad L_{new} = (0.4225 - 1)^2 = 0.33350625$$

The same parameter appears in both encoder and decoder. Differentiating only one occurrence loses a path through the computation. This is a toy tied-weight problem; independent encoder and decoder parameters would have separate derivatives.

D. Convolution output and parameter count [suggested 4 marks]

Question: a 32 x 32 x 3 input enters a convolution with 16 output channels, kernel 4 x 4, stride 2, padding 1, dilation 1, and one bias per output channel. Find the output shape and trainable parameter count.

$$n_{out} = \left\lfloor \frac{n + 2p - k}{s} \right\rfloor + 1 = \frac{32 + 2 - 4}{2} + 1 = 16$$

$$\text{output} = 16 \times 16 \times 16$$

$$\#\text{parameters} = (4 \cdot 4 \cdot 3 + 1)16 = 784$$

There are 48 kernel weights plus one bias for each of 16 filters. Spatial output positions reuse those weights, so the 16 x 16 output area does not multiply the parameter count.

Suggested marks: output rule and shape 2; weights plus biases 2. Shape exercises are a useful architecture check, with weaker direct evidence than the quiz's loss and metric questions.

Alternate E-F: conditioning and coverage

Optional: explanations that connect architecture to evaluation

E. cGAN and StackGAN [suggested 5 marks]

Question: write a class-conditioned GAN objective. Why should the discriminator see the condition? Explain the purpose of the two StackGAN stages.

$$\min_G \max_D \; \mathbb{E}_{(x,\,c)\sim p_{data}}[\ln D(x,c)] + \mathbb{E}_{c,\,z}[\ln(1 - D(G(z,c),c))]$$

The generator receives noise z and condition c; the discriminator judges real/fake compatibility jointly with c. If D never sees c, a generator may learn plausible images while ignoring the requested class because realism alone can satisfy its feedback.

StackGAN uses a first stage for a coarse, lower-resolution text-conditioned image and a second stage that uses the coarse image and text information to refine details at higher resolution. Coarse-to-fine construction makes the task easier to organize; it does not guarantee perfect correction or text alignment.

Suggested marks: objective 2; why both networks condition 1; roles of the two stages 2.

F. Interpret generative precision and recall [suggested 5 marks]

Question: use a fixed real reference set. For model A, 18 of 20 generated features lie inside the real feature manifold, and 6 of 20 real features lie inside A's generated manifold. For B, the corresponding counts are 12 and 17. Compute precision/recall and diagnose the trade-off.

Model	Precision: generated inside real	Recall: real inside generated
A	18/20 = 0.90	6/20 = 0.30
B	12/20 = 0.60	17/20 = 0.85

A emphasizes fidelity but covers little of the real variety, consistent with missing modes. B covers more real variety but includes more generated features outside the real manifold. Neither pair alone determines the best model for every application.

These are manifold-membership ratios, not label-classification precision and recall. Membership itself depends on the chosen feature extractor and neighborhood construction. The supplied counts avoid having to build the neighborhoods in this question.

A generator can memorize one convincing sample per class and receive a high IS. A single confident class, however, gives IS=1. Keep across-class and within-class diversity separate.

Alternate G-H: cross-attention and DDPM

Optional: one numerical transfer and one compact derivation

G. Compute one cross-attention output [suggested 5 marks]

Question: one decoder query is q=(1,0). Encoder keys are k_1=(1,0), k_2=(0,1); values are v_1=(2,0), v_2=(0,4). Use d_k=2. Find scores, weights, output and shapes.

$$s = (1/\sqrt{2}, 0) = (0.707107, 0)$$

$$a = (0.669762, 0.330238), \quad o = 0.669762(2, 0) + 0.330238(0, 4)$$

$$o = (1.339523, 1.320954)$$

Q is 1 x 2, K and V are 2 x 2, scores/weights are 1 x 2, and the output is 1 x 2. The query comes from the decoder; keys and values come from the encoder. Do not replace the value vectors with keys when forming the output.

H. Recover the DDPM reverse-mean formula [suggested 6 marks]

Question: starting from the exact posterior mean below, substitute the clean-image estimate implied by predicted noise. Show why the result equals the mean used in Q6.

$$\tilde{\mu} = Ax_0 + Bx_t, \quad A = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}, \quad B = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}$$

$$\hat{x}_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\,\varepsilon_\theta}{\sqrt{\bar{\alpha}_t}}$$

Because alpha_bar_t = alpha_t alpha_bar_(t-1), substitution gives these two coefficients:

$$\frac{A}{\sqrt{\bar{\alpha}_t}} + B = \frac{\beta_t + \alpha_t(1-\bar{\alpha}_{t-1})}{\sqrt{\alpha_t}(1-\bar{\alpha}_t)} = \frac{1}{\sqrt{\alpha_t}}$$

$$\frac{A\sqrt{1-\bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} = \frac{\beta_t}{\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t}}$$

$$\mu_\theta = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon_\theta\right)$$

The learned noise estimate replaces unavailable clean-image information. This does not make the estimate equal to the true x_0 for every sample.

Alternate I-J: explain the whole process

Optional: answer these aloud after the calculations

I. DDPM training, sampling and objective [suggested 6 marks]

Question: describe one DDPM training example and the generation loop. State the simple loss, and explain what changes when a weighted timestep loss is replaced by the unweighted simple loss.

Training: sample a clean data example x_0, a timestep t, and independent standard-normal noise epsilon. Compute x_t directly using the cumulative forward schedule. Pass x_t and t to the network; compare predicted noise with the actual aggregate noise used to construct x_t.

$$L_{simple} = \mathbb{E}_{x_0, t, \varepsilon}\left[\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2\right]$$

Sampling: start from x_T approximately standard normal, predict noise at each t, compute the reverse mean, add the chosen Gaussian noise for t>1, and repeat until x_0. Generation does not start from a known clean training image. The forward corruption is predefined; the network learns the reverse denoising behavior.

For equal fixed Gaussian covariances, a posterior-to-model KL is proportional to a squared mean difference, which can be rewritten as a weighted noise error. Removing timestep-dependent positive weights changes how timesteps are emphasized: it is a training-objective choice, not subtraction of an additive constant. The simple loss is not literally the entire variational bound.

J. Compare AE, VAE, GAN and DDPM [suggested 4 marks]

Question: explain each model's training signal and how it would produce an output. Give one trade-off per model.

Model	Training / generation	One trade-off
AE	Encode input, decode code; train reconstruction. Decode a supplied latent code.	Useful representations; no automatic calibrated sampling prior.
VAE	Likelihood reconstruction plus prior KL; sample a prior code and decode.	Organized latent sampling; outputs may be smoother under common likelihood choices.
GAN	Adversarial discriminator feedback; transform sampled noise with G.	Fast one-pass sampling; unstable training or mode collapse can occur.
DDPM	Learn a noise predictor on corrupted examples; iteratively denoise sampled noise.	Strong distribution modeling; many reverse steps can make sampling costly.

Final self-check: state the derivative variable, distinguish variance from standard deviation, name each sum/mean, track matrix shapes, and explain the result in one sentence.

Review discussion: the Astra tutor independently checked the numerical answers and challenged the draft. We increased transformer coverage with explicit CE/position questions, kept long derivations optional for timing, and clarified GAN logit derivatives, decoder masks, FID limitations, and DDPM loss weighting.